Disaster Aid Platform
Analytics Service
Operational reporting unit

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Strengthening the Coordination of Disaster Aid Operations

Quarterly platform analytics report

Report of the Analytics Service of the Disaster Aid Platform

Summary

The present report has been prepared in support of operational oversight of the Disaster Aid platform and covers the period Q2 2026 (snapshot at 2026-04-28 UTC). It provides a quantitative overview of the activity recorded in the production database, examines volunteer engagement and retention trends, presents the inventory of active predictive models, and formulates a series of recommendations.

During the reporting window the platform processed 125 citizen reports, of which 89 were verified by moderators or automated credibility scoring (71 per cent verification rate, moderate). The system routed 77 help requests, 51 of which reached resolution (66 per cent resolution rate). The observed mean response time stood at 22.0 minutes, broadly aligned with the 30-minute service-level objective.

Volunteer engagement remains a structural priority. 248 approved volunteer hours have been logged across 2 tracked organisations, while the churn predictor identifies 2 volunteers above the 70 per cent risk threshold. The average churn probability (53 per cent) calls for sustained retention work and informs the recommendations set out in chapter VI.

Finally, the platform currently runs 2 active predictive models, covering disaster spread forecasting, volunteer churn prediction and credibility scoring. A complete inventory and recent audit-log activity are provided in chapters IV and V.

TOTAL REPORTS **125**	VERIFICATION RATE **71%**	HELP REQUESTS **77**
RESOLUTION RATE **66%**	AVG RESPONSE **22.0 min**	ACTIVE ML MODELS **2**
VOLUNTEER HOURS **248**	AVG CHURN **53%**	HIGH-RISK VOLUNTEERS **2**

Source: PostgreSQL · disaster_aid · ai_service

I. Introduction

1. The Disaster Aid platform is an integrated emergency-response system that aggregates citizen reports, sensor telemetry and responder coordination into a single operational picture. The present report consolidates the analytics surfaced by the platform's **/analytics** dashboard and covers Q2 2026 (snapshot at 2026-04-28 UTC). It is intended for programme leads, operations managers and platform engineers.

2. The data underlying the report are queried directly from the production PostgreSQL instance through the read-only *data_loader* module of the AI service. No external sources are used; consequently the figures presented hereafter reflect the exact state of the database at the time of generation.

3. At the snapshot date the database held 6 alert records, 0 citizen incident reports, 0 sensor entries, 0 disaster-statistics rows and 0 volunteer-statistics rows. These tables are the foundation of every aggregation that follows.

4. The report is organised into six chapters. Chapter II reviews operational throughput and quality of moderation. Chapter III covers volunteer engagement and retention. Chapter IV inventories the active machine-learning stack. Chapter V summarises recent audit-log activity. Chapter VI consolidates findings and recommendations.

II. Operational Landscape

This chapter examines the volume and quality of the operational throughput recorded by the platform during the reporting period, starting from inbound citizen reports and ending with the resolution of help requests dispatched to responders.

Citizen reports and verification

5. Throughout the reporting window the platform ingested 125 citizen reports. Of those, 89 (71 per cent) successfully cleared the moderation pipeline, which combines manual review by partner organisations with automated credibility scoring. The remaining 36 reports were either flagged as unverified, rejected or are still awaiting review.

6. A verification rate of 71 per cent is considered **moderate**. Sustained verification quality is essential because the credibility weight assigned to each report directly conditions downstream dispatch decisions and the training signal of the credibility scorer. Any prolonged drop in this indicator should trigger a review of moderator staffing levels and of the credibility scorer's threshold.

Help requests and resolution outcomes

7. Citizens and field volunteers submitted 77 help requests during the period. By snapshot time, 51 of them (66 per cent) had been formally resolved, while 26 remained open or were closed without resolution. This indicator is closely correlated with responder capacity in each geofenced zone and with the matching optimiser used by the coordination module.

8. The mean response time across all incidents in the period was 22.0 minutes, a level broadly aligned with the 30-minute service-level objective. Response time is computed as the elapsed period between the moment an alert is acknowledged by an operator and the moment a responder is formally dispatched. Spikes in this indicator typically reflect either an under-supply of responders or notification fan-out delays affecting the websocket broker.

Coverage and underlying data volumes

9. The operational picture is built on top of 6 alert records, 0 sensor entries and 0 citizen incident reports. Where any of these volumes drops sharply between consecutive reports, the cause should be investigated as a matter of priority — sustained throughput is the precondition for every metric described in this chapter.

III. Volunteer Engagement and Retention

Volunteer activity is the structural input on which the platform depends. This chapter reviews the volume of validated contributions, situates the current churn-risk profile and highlights the cases that require organisational follow-up.

Engagement and approved hours

10. During the reporting window, 248 hours of volunteer time were approved across 66 opportunities and 2 tracked organisations. Approved hours represent the subset of logged hours that have passed organisational validation; they are the canonical input to the volunteer credibility scorer and to the gamification engine.

11. Hours are aggregated per volunteer in the *volunteer_statistics* table, which is recomputed at regular intervals by the analytics scheduler. Reviewing the distribution of these hours is a useful proxy for identifying both top contributors deserving recognition and low-engagement cohorts at risk of disengagement.

Churn risk and retention outlook

12. The churn predictor currently estimates an average churn probability of 53 per cent across the active volunteer base. A total of 2 volunteers exceeds the 70 per cent threshold and qualifies for a high-risk intervention. Such interventions can take the form of an opportunity recommendation, an inclusion in a team challenge, or an achievement nudge, depending on the volunteer's recent activity profile.

13. The most exposed individual case in the current snapshot presents a predicted churn probability of 91 per cent. Concentrated cases of this magnitude warrant proactive contact by the responsible organisation; the historical record indicates that early outreach within forty-eight hours of the prediction materially improves the reactivation rate.

IV. Predictive Analytics and Machine-Learning Stack

14. Predictive scoring is delivered by the models registered in the *predictive_model* table. Five model families are currently supported: spread predictor, matching optimiser, credibility scorer, churn predictor and impact estimator. A model is only considered authoritative for production traffic when its **is_active** flag is set.

15. At the time of writing, 2 models are flagged as active, covering the following families: *CREDIBILITY_SCORER, SPREAD_PREDICTOR*. The most recently trained active model is **Credibility Score V2** version 2.1.0, retrained on 2026-04-21 18:17:41.382515. An exhaustive list, including version identifiers and training timestamps, is provided in the appendix tables.

16. Each model is registered with an artefact reference, an optional metrics blob and an activation flag. Whenever a model is retrained, the previous version should be deactivated rather than deleted, in order to preserve a complete production lineage and to enable rollback in case of regression.

V. Recent Operational Activity

17. The audit log is currently empty. Although this is consistent with a freshly seeded environment, in production this state would warrant an immediate investigation: every configuration change, alert lifecycle transition and

machine-generated decision is expected to leave a trace in the *audit_log* table.

VI. Findings and Recommendations

This concluding chapter consolidates the principal findings of the analysis and formulates a series of recommendations. Each recommendation is anchored in the figures presented in the preceding chapters; their adoption is intended to maintain the platform's operational quality and to anticipate emerging risks.

20. Verification quality stands at 71 per cent, a level qualified as **moderate**. Action is required to bring this indicator above 75 per cent in the next reporting cycle.

21. Resolution rate stands at 66 per cent. The platform is therefore failing to close a material share of incoming requests; this is consistent either with insufficient responder supply or with bottlenecks in the matching optimiser.

22. The mean response time of 22.0 minutes is broadly aligned with the 30-minute service-level objective. No corrective action is needed beyond ongoing monitoring.

23. Volunteer churn risk averages 53 per cent. The platform is exposed to material attrition risk and should activate targeted retention interventions before the next reporting cycle.

a) **Operational.** Increase moderator capacity or recalibrate the credibility-scorer threshold so as to bring the verification rate above 75 per cent in the next reporting cycle.

b) **Volunteer engagement.** Trigger targeted retention interventions — opportunity recommendations, team challenges, achievement nudges — for every volunteer whose predicted churn probability exceeds forty per cent, with a particular focus on the high-risk cohort.

c) **Data quality.** Validate, ahead of every external publication of these analytics, that the disaster and volunteer statistics tables have been re-computed within the last twenty-four hours and that no aggregation job has silently failed.

d) **Reporting cadence.** Generate the present report at least once per quarter and circulate it to programme leads and platform engineers, so that operational drift can be detected and addressed in a timely manner.

Annex — Detailed metrics and tabular data

A. Disaster analytics — full KPI list

Total citizen reports	**125**
Verified reports	**89**
Verification rate	**71%**
Total help requests	**77**
Resolved requests	**51**
Resolution rate	**66%**
Avg response time (min)	**22.00**
Audit log entries	**2**
Active predictive models	**2**

B. Volunteer analytics — full KPI list

Total approved hours	**247.5**
Total opportunities	**66**
Average churn probability	**53%**
High-risk volunteers (> 70%)	**2**
Organisations tracked	**2**

C. Active predictive models

Name	Version	Type	Trained at
Credibility Score V2	2.1.0	CREDIBILITY_SCORER	2026-04-21 18:17
Spread V1	1.0.0	SPREAD_PREDICTOR	2026-04-15 18:17

E. High-risk volunteer churn

Volunteer	Churn	Intervention	Predicted at	Reactivated
2003	91%	ACHIEVEMENT_NUDGE	2026-04-27 18:07	no
2001	83%	TEAM_CHALLENGE	2026-04-27 17:22	no